

ROUTE R1: THE COMPLEX STOKES CONSTANT OF THE TW_β TAIL, COMPUTED VIA THE PAINLEVÉ II MEDIAN RESUMMATION

SOLOMON WILLIAMS

ABSTRACT. Route R1 closes the qualitative question of the last open piece — the complex Stokes constant on the $s \rightarrow -\infty$ branch — and computes the backbone value. By Theorem T1 the tail coefficients are Painlevé II trans-series coefficients, so the Stokes datum is inherited from the Hastings–McLeod solution, which we compute directly. The $s \rightarrow -\infty$ asymptotic coefficients of $q(s)$, $a_0 = 1$, $a_1 = -\frac{1}{8}$, $a_2 = -\frac{73}{128}, \dots$, are *non-sign-alternating* (all $a_{k \geq 1} < 0$) — the Borel singularity sits on the positive real axis, the lateral Borel sums differ by an imaginary amount, and the median is the real tail. This is the genuine median-resummation Stokes structure. The large-order growth is $a_k \sim -C \Gamma(2k - \frac{1}{2}) (9/8)^k$ with $C \approx 0.139 \approx 1/(4\sqrt{\pi})$, giving the Borel-singularity/action $A = \frac{2\sqrt{2}}{3}$ and a nonzero (imaginary-ambiguity) Stokes constant. It also corrects a framing slip in the O5c note: the tail Stokes datum is a single function inherited from PII, not an $O(1/\beta)$ “shift”.

1. THE COMPUTATION

By Theorem T1 the fluctuation coefficients $c_n(a)$ of the tail are Painlevé II trans-series coefficients; the median-resummation Stokes datum they carry is therefore that of the Hastings–McLeod solution on its $s \rightarrow -\infty$ branch. We compute the latter directly. Substituting $q(s) = \sqrt{-s/2} \sum_{k \geq 0} a_k (-s)^{-3k}$ into $q'' = 2q^3 + sq$ gives the recursion, solved exactly:

$$a_0 = 1, \quad a_1 = -\frac{1}{8}, \quad a_2 = -\frac{73}{128}, \quad a_3 = -\frac{10657}{1024}, \quad a_4 = -\frac{13912277}{32768}, \quad \dots \quad (1)$$

($a_1 = -\frac{1}{8}$ and $a_2 = -\frac{73}{128}$ are the known Hastings–McLeod values).

Proposition 1 (Median-resummation Stokes structure). *The coefficients (1) are non-sign-alternating: $a_k < 0$ for all $k \geq 1$. Hence the Borel transform has its singularity on the positive real axis, the lateral Borel sums $\mathcal{S}_\pm$ differ by a purely imaginary amount, and the physical (real) asymptotics is their median $\frac{1}{2}(\mathcal{S}_+ + \mathcal{S}_-)$. The large-order growth is*

$$a_k \sim -C \Gamma(2k - \frac{1}{2}) \left(\frac{9}{8}\right)^k, \quad C \approx 0.139 \left(\approx \frac{1}{4\sqrt{\pi}}\right), \quad (2)$$

verified numerically: $a_{k+1}/a_k/(4k^2) \rightarrow 1.126 \rightarrow \frac{9}{8}$, and C stable under Richardson extrapolation over $k \leq 9$. The Borel-singularity/action is $A = \frac{2\sqrt{2}}{3} \approx 0.943$ (in the variable $(-s)^{3/2}$), and the Stokes constant — the imaginary discontinuity across the positive-axis singularity, $\text{Im } \mathcal{S} \sim \pi C e^{-A(-s)^{3/2}}$ — is nonzero. [computed]

Corollary 2 (The TW_β tail inherits this). *By Theorem T1 the tail trans-series carries exactly this median-resummation Stokes structure: a positive-axis Borel singularity, an imaginary lateral ambiguity, and a nonzero Stokes constant of Hastings–McLeod type. So the complex Stokes datum of TW_β — the genuine analogue of the cusp’s complex structure — exists and is computed at the backbone level, and is a **trans-series** (median-resummation) phenomenon, not a resonance shift (O5c). [computed]/[cited]*

Date: July 21, 2026.

2. CORRECTION TO O5C, AND WHAT REMAINS

The O5c note wrote the Stokes constant as $S(\beta) = S_{\text{HM}} + \beta^{-1}S_1 + \dots$. That decomposition is imprecise: the Stokes datum is a single function $S(a)$ read from the large-order growth of the $c_n(a)$ (not itself a $1/\beta$ series), and by Theorem T1 its *structure* and *backbone value* are the Hastings–McLeod ones of Proposition 1. There is no separate “ $O(1/\beta)$ noise shift” of the Stokes constant; the $1/\beta$ is the trans-series variable whose Stokes structure is PII’s.

What is closed: the qualitative question — *is* there a complex Stokes constant for TW_β , and of what type — is answered: yes, a median-resummation (imaginary-ambiguity) Stokes constant, computed via the Hastings–McLeod backbone (Prop. 1, Cor. 2), with action $\frac{2\sqrt{2}}{3}$, index $-\frac{1}{2}$, amplitude ≈ 0.139 . *Refinement remaining:* the precise a -dependence $S(a)$ of the tail-specific Stokes constant (as opposed to the backbone value) would require the tail coefficients $c_n(a)$ themselves — the transport hierarchy of the O1 note — and an independent value of C (e.g. matching the exact PII connection constant of Its–Kapaev, to test $C \stackrel{?}{=} 1/(4\sqrt{\pi})$). [refinement]

REFERENCES

- [1] N. J. Cleri, G. V. Dunne, *Resurgent trans-series for generalized Hastings–McLeod solutions*, J. Phys. A **53** (2020) 305201; arXiv:2002.06270.
- [2] A. S. Fokas, A. R. Its, A. A. Kapaev, V. Yu. Novokshenov, *Painlevé Transcendents: The Riemann–Hilbert Approach*, AMS Math. Surveys Monogr. **128**, 2006.